

YUHANG JIANG

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Summary

- EIT Manufacturing double-degree master's in Computer Science (UNITN) and Data Science (SUPSI).
- Two first-author journal papers accepted in 2026: IEEE Transactions on Geoscience and Remote Sensing and TMLR.
- GeoSelect sets the training-free state of the art on both RRSIS-D and RISBench, reaching about 93% of a supervised specialist on RRSIS-D.
- Research intern at Huawei Pisa Research Center and FBK. Kaggle Competitions Expert.
- Work spans embodied AI, mechanistic interpretability of state-space models, adversarial robustness, and remote sensing.

Education

EIT Manufacturing Master & Doctoral School (Double Degree)

Sep 2023 - Mar 2026

Data Science and AI for a Competitive Manufacturing

- **Università di Trento (UNITN), Italy**
MSc, Computer Science · GPA 98/110
- **University of Applied Sciences and Arts of Southern Switzerland (SUPSI), Switzerland**
Master of Engineering (MSE), Data Science · GPA 5.1/6.0
- **Scholarship:** Recipient of the EIT Manufacturing Scholarship (full tuition waiver + 10,000 EUR/yr)

Anhui University (Project 211), China

Sep 2019 - Jun 2023

BEng, Intelligent Science and Technology

- **Average Score:** 84.34/100
- **Awards:** Second Prize for Outstanding Academic Performance (2022); First Prize Scholarship of Academic Science and Technology (2021)
- **Courses:** Machine Learning, Pattern Recognition, Big Data Analysis, Numerical Analysis, Natural Language Processing.

Work Experience

Huawei Pisa Research Center

Apr 2026 - Oct 2026

Research Intern

Pisa, Italy

- **Led an independent research project on memory for multimodal agents**, from problem definition and experiment design to paper; first-author paper under review. Controlled experiments that hold the retrieved set fixed and change one variable at a time located the bottleneck in **delivery** rather than retrieval: delivering the original images is worth +13.87 points, perfect retrieval only +2.31, and the delivery gain grows on a stronger backbone (+17.92 at 32B).
- **Proposed DeliverMem, a training-free delivery method that leaves the memory store unchanged** (original modality, readable identifiers and temporal position, plus a retrieval-side adaptation): **state of the art on DMV-Bench**, ahead of the best prior method at every setting on both backbones by 7.63 to 22.62 points; on MemLens it leads the strongest published memory-augmented agent by at least 8.20 points at all four context lengths, on a tenth to a seventieth of the full-context input. At MemLens 32K with dev-fitted selection it **surpasses GPT-5.4 and matches Gemini-3.1-Pro** while delivering about 2,400 tokens, a tenth of what they read.
- **Owned the LLM post-training pipeline for cockpit AI end to end:** data construction and cleaning, SFT, RL with reward design, evaluation and result analysis.
- Built inference, training and evaluation pipelines on vLLM, Transformers and ms-swift, supporting large-scale controlled comparisons and ablations across backbones and context lengths.

Fondazione Bruno Kessler - FBK

Mar 2025 - Mar 2026

Research Intern

Trentino-Alto Adige, Italy

- Proposed a new embodied perception task and built **PinVerify**, the first offline multi-view benchmark for active instance verification in realistic 3D environments.
- Designed an MLLM-based agent with attribute decomposition, confidence-weighted multi-view tracking, and next-best-view planning; fine-tuned via LoRA with SFT followed by RL alignment, achieving 85.6% accuracy.

- Researched the latest machine learning and industrial anomaly detection algorithms.
- Applied GAN models to reconstruct images to identify anomalies and evaluate the performance.
- Used the object detection neural network model to detect abnormal parts in high-speed rail supports.
- Assisted colleagues in developing a set of character recognition patents for LCD meters.

Publications

"GeoSelect: Spatial-Program Execution for Training-Free Referring Remote Sensing Image Segmentation"
 IEEE Transactions on Geoscience and Remote Sensing (TGRS), 2026, [IEEE Xplore](#) · [Project Page](#),
[arXiv:2607.03869](#)

Reframes referring expression resolution as the execution of an explicit spatial program, covering the positional superlative, ordinal, and compositional expressions that continuous fields cannot. Reaches 58.86 mIoU on RRSIS-D and 55.27 on RISBench, more than twice and about 1.7 times the best prior training-free results.

Yuhang Jiang, Guohui Deng*, Miao Zhong Xu, Chao Ruan, Jinling Zhao, Linsheng Huang

"The Scissors Effect: When Resize-Based Input Diversity Helps or Hurts Transfer Attacks"

Transactions on Machine Learning Research (TMLR), 2026, [Project Page](#), [arXiv:2606.22516](#), [OpenReview](#)

Shows that resize-based input diversity, applied by default in transfer attacks, reverses sign on adversarially trained surrogates. Robustness evaluations that apply it blindly are therefore optimistic about the models they test.

Yuhang Jiang, Xiaojing Chen

"PlnVerify: An Offline Embodied Benchmark for Active Instance Verification"

Poster, FMEA Workshop @ CVPR 2026, [Project Page](#), [arXiv:2605.30639](#), [OpenReview](#)

Master's thesis project.

Proposes Active Instance Verification, a new embodied task, and builds the first benchmark for it. Finds that active viewpoint selection gives no reliable gain over choosing at random, and that detection quality, not verification logic, is what binds the task.

Yuhang Jiang

Note: * indicates the corresponding author.

Preprints and Other Publications

"A Circuit, Not The Circuit: Non-Unique Causal Localisation of the Mamba-2 State Sink"

[arXiv:2606.00930](#)

Names the state sink, the state-space analogue of the attention sink. A near-disjoint set of heads matches the causal effect of the probe-selected specialists, and the effect dissociates by intervention surface, so a probe recovers a circuit rather than the circuit.

Yuhang Jiang, Bowen Zhang

"Refit the Probe: Single-Direction Ablation Is Not a Necessity Test"

[arXiv:2606.00926](#)

Ablating the direction a probe finds does not remove the target: a probe refitted on the ablated activations recovers its original accuracy, and keeps recovering when the probe's entire row space is deleted, because the quantity survives in the orthogonal complement. Neither a large task drop nor a near-zero one therefore establishes necessity. Replacing the ablation with iterative nullspace projection, scored against random subspaces of matched dimension, reverses the conclusion: representations that looked causally inert carry most of the task.

Yuhang Jiang

"CL-RAG: A Closed-Loop Multimodal Retrieval-Augmented Generation Architecture for Robust Human-Robot Control Interaction"

WRC SARA 2025 (oral) · Bowen Zhang, Yuhang Jiang, Lingxiang Hu, Dun Li, Qianqian Hu*

"Improved lightweight identification of agricultural diseases based on MobileNetV3"

CAIBDA 2022 (oral) · Yuhang Jiang*, Wenping Tong

Software Copyright, Intelligent contract software for agricultural insurance compensation, 2022SR1568111.

Academic Service

Reviewer, ICLR 2027.

Reviewer, InterpScience (Interpretability as a Science) Workshop @ NeurIPS 2026.

Reviewer, TTCL (Test-Time Continual Learning) Workshop @ NeurIPS 2026.

Reviewer, FMEA (Foundation Models Meet Embodied Agents) Workshop @ CVPR 2026.

Selected Projects

Research on Semantic Segmentation Method of High-Resolution Remote Sensing Images May 2023
Outstanding Undergraduate Graduation Project

- Developed an encoder-decoder model with residual-weighted attention for remote sensing segmentation, achieving F1-scores of 90.23% and 87.37% on ISPRS Potsdam and Vaihingen datasets.

Other projects. Multi-objective flexible job-shop scheduling with an enhanced NSGA-II (2025); HGT-GWO, a parallel Grey Wolf Optimizer in C/MPI/OpenMP on UNITN's HPC cluster (2025); AR-based robotic arm control on HoloLens 2 with YOLOv8 and FastSAM (2024).

Competitions

Kaggle March Machine Learning Mania 2025 - Silver Medal (64th / 1,727) Apr 2025
Solo

Kaggle BirdCLEF 2024 Competition - Bronze Medal (70th / 974) Jun 2024
Solo

Kaggle BirdCLEF+ 2025 Competition - Bronze Medal (199th / 2,025) Jun 2025
Solo

Mathematical Contest In Modeling (MCM), Finalist winner (top 1%) Apr 2021
Leader

- Designed a set of algorithms to assess the degree of hunger around the world and optimize the food supply chain.
- Responsible for modeling, programming and part of paper writing.
- Youngest winner in the history of Anhui University; first winner from the School of Internet.

iCAN Innovation Contest 2021 (Anhui) - Second prize Oct 2021
Team member

- Developed a set of hardware system capable of intelligently monitoring and warning the abnormal behavior of the elderly.
- Responsible for some STM32 development and project plan writing.

Skills

- Programming:** Python, C/C++, Matlab, R.
- ML / DL Frameworks:** PyTorch, scikit-learn; MPI / OpenMP (HPC).
- LLM / VLM:** multimodal LLMs (MLLMs), post-training (SFT, DPO, GRPO/GSPO), LoRA / PEFT, RAG, LLM agents.
- Architectures:** Transformer, Mamba, CNN, RNN.
- Applications:** Image Classification & Segmentation, Industrial Anomaly Detection, Time-Series Analysis.
- Other:** Unity; Lean Manufacturing & Sustainable Management.

Language

- English:** Fluent, TOEFL 102
- Chinese:** Native Speaker